

# The use of a natural coagulant (*Opuntia ficus-indica*) in the removal for organic materials of textile effluents

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**Abstract** The goal of this study was to investigate the activity of the coagulant extracted from the cactus *Opuntia ficus-indica* (OFI) in the process of coagulation/flocculation of textile effluents. Preliminary tests of a kaolinite suspension achieved maximum turbidity removal of 95 % using an NaCl extraction solution. Optimization assays were conducted with actual effluents using the response surface methodology (RSM) based on the Box–Behnken experimental design. The responses of the variables  $\text{FeCl}_3$ , dosage, cactus dosage, and pH in the removal of COD and turbidity from both effluents were investigated. The optimum conditions determined for jeans washing laundry effluent were the following:  $\text{FeCl}_3$  160 mg L<sup>-1</sup>, cactus dosage 2.60 mg L<sup>-1</sup>, and pH 5.0. For the fabric dyeing effluent, the optimum conditions were the following:  $\text{FeCl}_3$  640 mg L<sup>-1</sup>, cactus dosage 160 mg L<sup>-1</sup>, and pH 6.0. Investigation of the effects of the storage time and temperature of the cactus *O. ficus-indica* showed that coagulation efficiency was not significantly affected for storage at room temperature for up to 4 days.

**Keywords** Coagulation/flocculation · Natural coagulant · Textile effluents · Box–Behnken

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## Introduction

Industrial activities are some of the greatest causes of contamination of hydric resources, because most industrial processes use large amounts of water and produce effluents that contain toxic species or species that are difficult to degrade (Garcia et al. 2007; Senthilkumar et al. 2011a). Among the industrial processes responsible for the generation of effluents stands out the textile sector, whose effluents are characterized by strong color and great amounts of chemical compounds that are toxic to human beings and the environment. Many textile dyes and by-products are carcinogenic and/or mutagenic (Garcia et al. 2009a, b; Pinheiro et al. 2004; Senthilkumar et al. 2011b).

The coagulation/flocculation technique followed by sedimentation is used worldwide in the treatment of residual waters (Aber et al. 2010; Khayet et al. 2011; Prasad 2009). This treatment consists in the quick dispersion of the coagulating agent in the water to be treated, followed by intense agitation, which is commonly defined as quick mixing (Rossini et al. 1999). Coagulants are added to effluents to destabilize the colloidal material and aggregate the small particles to form greater species, which are more easily removed as flocs (Rodrigues et al. 2008).

Aluminum salts are the most used inorganic coagulants due to their good performance, high efficiency, low cost, easy handling, and large availability (Bhatia et al. 2007). However, recently, the use of these salts has been questioned due to environmental impact of the large amount of sludge produced and the high level of

concentration of remaining aluminum in treated water. Various studies have associated high levels of ingestion of aluminum with Alzheimer's disease (Devrimci et al. 2012; Divakaran and Pillai 2002; Yin 2010; Simionato et al. 2006; Stephenson and Duff 1996; Zhao et al. 2012). Ferric salts and synthetic polymers have been used as an alternative for aluminum; however, their long-term environmental impact has not been fully investigated (Letterman and Pero 1990; Ndabigengesere et al. 1995). In the recent years, the growing world concern with environmental issues has raised the interest in the research of natural polyelectrolytes because they do not have great toxicological effects and of their greater biodegradability, safety to human health, and consequently, smaller environmental impact than metals or synthetic polymers (Bhatia et al. 2007; Prasad 2009; Zhang et al. 2006). Furthermore, the use of natural polyelectrolytes can reduce effluent treatment costs, as they can be grown, cropped, and processed locally (Sanghi et al. 2006). Cactus *Opuntia ficus-indica* (OFI) belongs to the Cactaceae species and can be easily grown in any region of Brazil. Studies indicate that OFI has a great potential as a natural coagulant (Diaz et al. 1999; Miller et al. 2008; Zhang et al. 2006). The goal of this study was to investigate the combined use of inorganic coagulant,  $\text{FeCl}_3$ , and aided coagulation with OFI in the treatment of actual textile effluents from a jeans laundry (JL) and fabric dyeing mill (FDM) by coagulation/flocculation process. The response surface method (RSM) was used to determine the influence of individual factors, namely, coagulant dosage, aided coagulation, and pH, and the influence of their interactions.

## Material and methods

The textile effluents investigated were obtained from a jeans laundry (JL) and a fabric dyeing mesh (FDM), both located in Northwest Paraná state, Brazil. The reagents used were analytical grade without previous purification: NaCl, KCl and  $\text{NaNO}_3$  (Synth), kaolin and  $\text{FeCl}_3 \cdot 6\text{H}_2\text{O}$  (FMaia).

### Chemical analysis

The moisture and ash contents of the OFI were determined using the method described by AOAC, and the crude protein content was determined by the Kjeldahl

method (Cunnif 1998). The total lipid (TL) of the OFI was extracted with a chloroform and methanol mixture (2:1 v/v), according to Folch et al. (1957).

### Effect of salt solution on coagulation activity

Samples of OFI obtained in South Brazil were washed and manually peeled. Only the cactus pulp (gum) was used. The gum obtained was stored in polyethylene bags and frozen. From the substrate obtained in this first step, the *in natura* and lyophilized active component was obtained following Okuda et al. (1999). The extraction solutions ( $1.0 \text{ mol L}^{-1}$ ) used were sodium chloride (NaCl), potassium chloride (KCl), and sodium nitrate ( $\text{NaNO}_3$ ). In the extraction process, 1.0 g of OFI was added to 100 mL of the extraction solution and agitated for 10 min in a magnetic stirrer. The resulting suspension was filtrated with Whatman 42 filter paper.

Lyophilized samples of the natural polyelectrolyte coagulant were tested for activity stability.

### Preparation of synthetic turbid water

In all coagulation experiments, turbid water samples were prepared by adding kaolin to water. Ten grams of kaolin was added to 1.0-L water. This suspension was stirred for 1 h for uniform dispersion of the kaolin particles and then stood for 24 h to allow for complete hydration of the particles. This suspension was used as a stock solution (Okuda et al. 1999).

### Coagulation/flocculation process

The coagulation/flocculation process was done with coagulant  $\text{FeCl}_3$ ,  $0.37 \text{ mol L}^{-1}$  stock solution, and 1 % (w/v) OFI as a coagulation aid dissolved in  $1.0 \text{ mol L}^{-1}$  NaCl. The assays were conducted in a jar test apparatus (microcontrolled model Millan-JT 203/6) with 250 mL of effluents previously quick agitated for 30 s (120 rpm), followed by 15 min slow agitation (20 rpm) (Rodrigues et al. 2008). The coagulant and coagulation aid were added before agitation. After 30-min stabilization, the supernatant of each flask was removed for determination of turbidity in a turbidimeter (Policontrol). The chemical oxygen demand (COD) was determined by the colorimetric method (Greenberg et al. 1992).

## Response surface methodology

RSM is a collection of mathematical and statistical techniques that are useful for the modeling and analysis of problems in which a response of interest is influenced by several variables, and the objective is to optimize this response. RSM also quantifies relationships among one or more measured responses and the vital input factors (Noordin et al. 2004).

The Box–Behnken design is an independent quadratic design in that it does not contain an embedded factorial or fractional factorial design. In this design, the treatment combinations are at the midpoints of edges of the process space and at the center. These designs are rotatable (or near rotatable) and require three levels of each factor. Box–Behnken design usually works well for process optimization (Islam and Lye 2009; Kousha et al. 2012).

The independent variables (factors) used in this experiment were coagulant concentration ( $\text{FeCl}_3$ ), coagulation aid concentration (OFI), and pH, for both the JL and the FDM effluents (Table 1). Three levels of independent variables were used: low (−1), central (0), and high (1). The levels were determined in previous studies as univariate analyses (Rodrigues et al. 2008) where the central point (0) on RSM was established as the optimal condition in this first studies. The parameters studied in the evaluation of the efficiency of the coagulation/flocculation process were the percentage of removal of turbidity and percentage of removal of COD for both effluents (Table 2).

**Table 1** Coded and actual levels of the independent factors used in the experimental design

| Factor                       | Symbol | −1  | Coded value<br>0<br>Actual value | 1   |
|------------------------------|--------|-----|----------------------------------|-----|
| EL                           |        |     |                                  |     |
| Coagulant dose <sup>a</sup>  | $X_1$  | 80  | 120                              | 160 |
| Flocculant dose <sup>a</sup> | $X_2$  | 1.6 | 2.4                              | 4.0 |
| pH                           | $X_3$  | 5.0 | 6.0                              | 7.0 |
| FDM                          |        |     |                                  |     |
| Coagulant dose <sup>a</sup>  | $X_1$  | 480 | 560                              | 640 |
| Flocculant dose <sup>a</sup> | $X_2$  | 120 | 140                              | 160 |
| pH                           | $X_3$  | 5.0 | 6.0                              | 7.0 |

<sup>a</sup> Concentration ( $\text{mg L}^{-1}$ )

The coded values of each level were obtained by Eqs. (1) and (2) and shows the second-order polynomial regression model used:

$$x_i = \frac{X_1 - X_0}{X_i} \quad (1)$$

$$y = \beta_0 + \sum_{i=1}^a \beta_i x_i + \sum_{i=1}^a \beta_{ii} x_i^2 + \sum_{i=1}^2 \sum_{j=i+1}^a \beta_{ij} x_i x_j + \varepsilon \quad (2)$$

where, in Eq. (1),  $x_i$  is the coded value for each factor,  $X_1$  is the actual value of each factor,  $X_0$  is the actual value of each factor at the central point, and ( $X_i$  is the difference between the levels of each factor. In Eq. (2),  $\beta_0$ ,  $\beta_i$ ,  $\beta_{ii}$ , and  $\beta_{ij}$  are the regression coefficients ( $\beta_0$  is the constant term,  $\beta_i$  is the linear effect term,  $\beta_{ii}$  is the quadratic effect term, and  $\beta_{ij}$  is the interaction effect term), and  $y$  is the response value predicted by the model. The statistical significance was evaluated by variance analysis (ANOVA) and also by the determination coefficient ( $R^2$ ). The statistical significance of the regression was evaluated by the  $F$  test, in which the ratio of the mean square of the regression and the residue mean square ( $F_{\text{calc}}$ ), in its respective degrees of freedom, was compared with the tabulated value of the  $F$  distribution at a 95 % confidence level. Similarly, the lack of fit was evaluated by comparing the ratio of the mean square of the lack of fit and the mean square of pure error ( $F_{\text{tab}}$ ), in their respective degrees of freedom, with the tabulated  $F$  value at 95 % confidence level. The  $p$  test also was used to evaluate the significance of the parameters for the model and the lack of fit. The result is significant if the  $p$  value is lower than 0.05 and opposite to the lack of fit. Insignificant lack of fit is desired as significant lack of fit indicates that there might be contributions in the regressor–response relationship that are not accounted for by the model (Myers and Montgomery 2002; Noordin et al. 2004). All of the experimental data were compiled using the Design Expert (version 7.1.3) software, from the elaboration of the experimental design, elaboration of the mathematical model, variance analysis, to the optimisation parameters.

## Results and discussion

### Effect of salt solution on coagulation activity and chemical analysis

The OFI active components were extracted using NaCl, KCl, and  $\text{NaNO}_3$ ,  $1.0 \text{ mol L}^{-1}$  each. Figure 1 depicts the

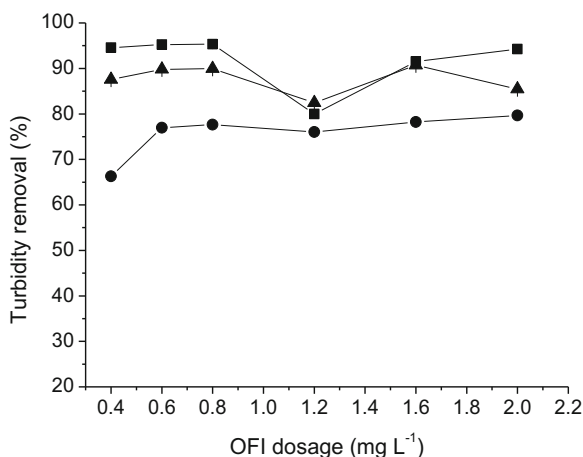
**Table 2** Box–Behnken design used for RSM with experimental values for the independent variables

| Run | JL              |                 |                 |                              |                                    | FDM             |                 |                 |                              |                                    |
|-----|-----------------|-----------------|-----------------|------------------------------|------------------------------------|-----------------|-----------------|-----------------|------------------------------|------------------------------------|
|     | $X_1^a (X_1)^b$ | $X_2^a (X_2)^b$ | $X_3^a (X_3)^b$ | COD removal (%) <sup>a</sup> | Turbidity removal (%) <sup>a</sup> | $X_1^a (X_1)^b$ | $X_2^a (X_2)^b$ | $X_3^a (X_3)^b$ | COD removal (%) <sup>a</sup> | Turbidity removal (%) <sup>a</sup> |
| 1   | 80.0 (−1)       | 1.60 (−1)       | 6.0 (0)         | 47.3964                      | 83.3981                            | 480.0 (−1)      | 120.0 (−1)      | 6.0 (0)         | 86.0367                      | 91.9355                            |
| 2   | 160.0 (+1)      | 1.60 (−1)       | 6.0 (0)         | 55.5460                      | 97.7670                            | 640.0 (+1)      | 120.0 (−1)      | 6.0 (0)         | 83.5054                      | 91.9355                            |
| 3   | 80.0 (−1)       | 4.00 (+1)       | 6.0 (0)         | 46.9675                      | 76.6990                            | 480.0 (−1)      | 160.0 (+1)      | 6.0 (0)         | 85.4416                      | 88.0645                            |
| 4   | 160.0 (+1)      | 4.00 (+1)       | 6.0 (0)         | 54.6167                      | 97.1845                            | 640.0 (+1)      | 160.0 (+1)      | 6.0 (0)         | 87.1926                      | 93.6129                            |
| 5   | 80.0 (−1)       | 2.80 (0)        | 5.0 (−1)        | 58.9774                      | 96.2136                            | 480.0 (−1)      | 140.0 (0)       | 5.0 (−1)        | 82.011                       | 72.5806                            |
| 6   | 160.0 (+1)      | 2.80 (0)        | 5.0 (−1)        | 60.4787                      | 99.0000                            | 640.0 (+1)      | 140.0 (0)       | 5.0 (−1)        | 86.3233                      | 82.2581                            |
| 7   | 80.0 (−1)       | 2.80 (0)        | 7.0 (+1)        | 45.6807                      | 64.0777                            | 480.0 (−1)      | 140.0 (0)       | 7.0 (+1)        | 88.6157                      | 75.8065                            |
| 8   | 160.0 (+1)      | 2.80 (0)        | 7.0 (+1)        | 64.7679                      | 91.2621                            | 640.0 (+1)      | 140.0 (0)       | 7.0 (+1)        | 87.8222                      | 84.1935                            |
| 9   | 120.0 (0)       | 1.60 (−1)       | 5.0 (−1)        | 57.9051                      | 98.3981                            | 560.0 (0)       | 120.0 (−1)      | 5.0 (−1)        | 77.9473                      | 76.7742                            |
| 10  | 120.0 (0)       | 4.00 (+1)       | 5.0 (−1)        | 52.6150                      | 98.8641                            | 560.0 (0)       | 160.0 (+1)      | 5.0 (−1)        | 77.8979                      | 75.1613                            |
| 11  | 120.0 (0)       | 1.60 (−1)       | 7.0 (+1)        | 43.4646                      | 75.1312                            | 560.0 (0)       | 120.0 (−1)      | 7.0 (+1)        | 80.8427                      | 74.1935                            |
| 12  | 120.0 (0)       | 4.00 (+1)       | 7.0 (+1)        | 49.9700                      | 74.7573                            | 560.0 (0)       | 160.0 (+1)      | 7.0 (+1)        | 86.1434                      | 78.0645                            |
| 13  | 120.0 (0)       | 2.80 (0)        | 6.0 (0)         | 59.4779                      | 96.7961                            | 560.0 (0)       | 140.0 (0)       | 6.0 (0)         | 80.8427                      | 92.9032                            |
| 14  | 120.0 (0)       | 2.80 (0)        | 6.0 (0)         | 61.7655                      | 95.1456                            | 560.0 (0)       | 140.0 (0)       | 6.0 (0)         | 79.6648                      | 94.8065                            |
| 15  | 120.0 (0)       | 2.80 (0)        | 6.0 (0)         | 56.8328                      | 94.1748                            | 560.0 (0)       | 140.0 (0)       | 6.0 (0)         | 79.4902                      | 93.7097                            |

<sup>a</sup> Actual values<sup>b</sup> Coded value

turbidity removal of the kaolinite turbidity effluent using the salts as extractors after the addition of the different concentrations of the OFI coagulant.

Figure 1 shows that all the investigated salts were efficient in the extraction of the OFI active component; the minimum turbidity removal was 65 %, and the maximum removal was 95 %. Similar results were obtained in a study



**Fig. 1** Comparison of the percent reduction of turbidity obtained for the extraction with salts: (black circle) NaCl, (black square) KCl, and (black triangle) NaNO<sub>3</sub> with OFI as a coagulation aid

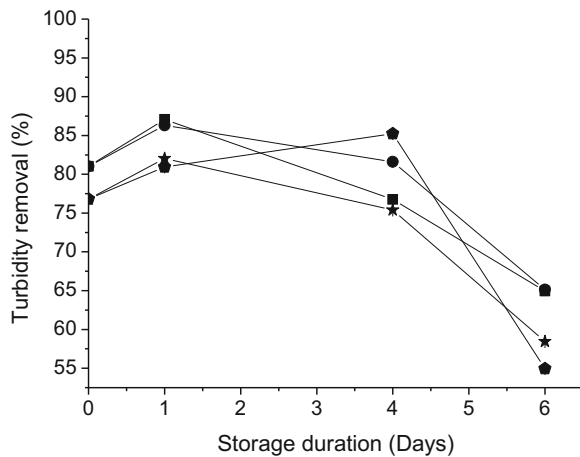
on saline extraction with *Moringa oleifera* seed (Okuda et al. 1999). According to Fig. 1, NaCl was more efficient in the removal of turbidity of the kaolinite effluent; thus, this salt was used in the study of other extraction processes of the OFI active component. The efficiency of these salts in the extraction of natural polyelectrolytes is related to the increase in the solubility of the protein responsible for the coagulant activity. The addition of a neutral salt to a species may increase its protein solubility (salting-in effect) due to increased solvation (Okuda et al. 1999).

Recent studies reported that galacturonic acid may be the active component that produces the coagulation capacity of OFI (Miller et al. 2008).

The analysis of chemical composition was performed in triplicate. The moisture content of the sample cactus OFI was 94.84 %±0.10. The percentages of ash, crude protein, and total lipids were the following: 1.42 %±0.050, 3.52 %±0.21, 0.28 %±0.015, respectively.

Effect of the natural polyelectrolyte storage time and temperature on coagulation efficiency

The stability of the coagulant activity of OFI after saline extraction (NaCl) was assessed for increasing storage



**Fig. 2** Percent reduction of turbidity for the variation of the extraction time with: (black square) *in natura* OFI, (black circle) *in natura* OFI stored in refrigerated environment, (black star) lyophilized OFI, and (black pentagon) lyophilized and refrigeration-stored OFI with NaCl as extraction solution

times of the extract after extraction. This study was conducted with *in natura* and lyophilized OFI extraction solutions which had been stored for 6 days at room temperature (28 °C) and under refrigeration (5 °C). Figure 2 presents the study results. One can see that the coagulant activity did not vary significantly after 4-day storage of the extraction solution.

Another important factor is that the coagulant activity of both the *in natura* and the lyophilized coagulant were similar. Therefore, only *in natura* OFI was used in the following studies.

#### Model responses and variance analysis (ANOVA)

The experimental design, the efficiency response in the removal of COD, and turbidity for JL and FDM are given in Table 2. The ranges of removal of COD of both effluents were 43.46–64.77 % and 77.90–88.62 %,

**Table 3** ANOVA statistical parameters obtained for the reduced models

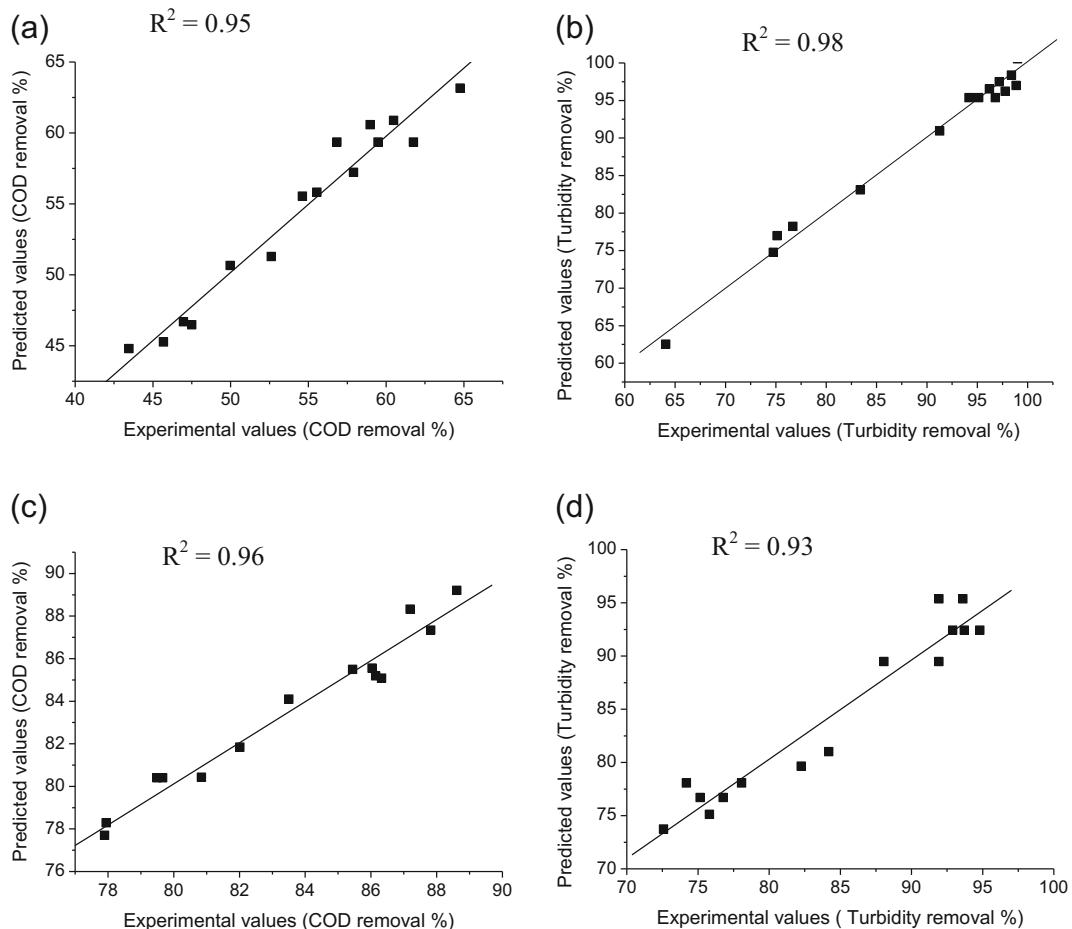
| Effluent | Source            | Sum of squares | DF | Mean square | $F_{calc}$ | $F_{0.05}$ | $P$ value | $R^2$ | CV (%) |
|----------|-------------------|----------------|----|-------------|------------|------------|-----------|-------|--------|
| JL       | COD removal       |                |    |             |            |            |           |       |        |
|          | Model             | 557.48         | 6  | 92.91       | 24.6       | 3.58       | <0.0001   | 0.95  | 3.57   |
|          | Residual          | 30.22          | 8  | 3.78        |            |            |           |       |        |
|          | Lack of fit       | 18.03          | 6  | 3.01        | 0.49       | 19.3       | 0.7875    |       |        |
|          | Pure error        | 12.19          | 2  | 6.09        |            |            |           |       |        |
|          | Total             | 587.70         | 14 |             |            |            |           |       |        |
|          | Turbidity removal |                |    |             |            |            |           |       |        |
|          | Model             | 1780.12        | 7  | 254.30      |            |            |           |       |        |
|          | Residual          | 29.85          | 7  | 4.26        | 59.6       | 3.79       | <0.0001   | 0.98  | 2.31   |
|          | Lack of fit       | 26.34          | 5  | 5.27        | 3.00       | 19.3       | 0.7875    |       |        |
|          | Pure error        | 3.51           | 2  | 1.76        |            |            |           |       |        |
|          | Total             | 1809.97        | 14 |             |            |            |           |       |        |
|          | COD removal       |                |    |             |            |            |           |       |        |
|          | Model             | 185.67         | 7  | 26.52       | 27.2       | 3.79       | 0.0001    | 0.96  | 1.19   |
|          | Residual          | 6.83           | 7  | 0.98        |            |            |           |       |        |
|          | Lack of fit       | 5.75           | 5  | 1.15        | 2.12       | 19.3       | 0.3503    |       |        |
|          | Pure error        | 1.08           | 2  | 0.54        |            |            |           |       |        |
|          | Total             | 192.50         | 14 |             |            |            |           |       |        |
| FDM      | Turbidity removal |                |    |             |            |            |           |       |        |
|          | Model             | 918.50         | 3  | 306.17      | 50.5       | 3.59       | <0.0001   | 0.93  | 2.92   |
|          | Residual          | 66.64          | 11 | 6.06        |            |            |           |       |        |
|          | Lack of fit       | 64.82          | 9  | 7.20        | 7.89       | 19.3       | 0.1175    |       |        |
|          | Pure error        | 1.83           | 2  | 0.91        |            |            |           |       |        |
|          | Total             | 985.14         | 4  |             |            |            |           |       |        |

CV (%) coefficient of variance, DF degree of freedom,  $R^2$  determination coefficient

respectively. However, we must point out that the concentrations of  $\text{FeCl}_3$  and OFI used in the optimization of the process with FDM were 4 and 40 times higher, respectively, than those used with JL. The increase in the concentration of the coagulant solution and the coagulation aid were necessary because the organic load added to this type of effluent is higher than that added to the JL due to the presence of dyes not fixed to the fibers during the dyeing process (about 20 % of the initial concentration used by industries) (Dallago and Smaniotto 2005; Guaratini and Zanoni 2000). Furthermore, these effluents are more concentrated than those from the laundry jeans washing process.

The statistical parameters obtained with variance analysis (ANOVA) for the model are presented in Table 3. The significance of the terms of the model was evaluated by  $F$  test with  $p$  at 95 % confidence level.

According to the results in Table 3, the four models were significant; the values of  $F_{\text{calc}}$  were greater than those of  $F_{0.05}$  and of  $p < 0.05$  at 95 % confidence level, which is desirable as it indicates that the terms in the model have a significant effect on the response (Noordin et al. 2004). Additionally, the models fitted well, since the  $F_{\text{cal}}$  values were smaller than those of  $F_{0.05}$  and than the values of  $p > 0.05$  at 95 % confidence level. The coefficient of variability (CV) as the ratio of the standard error of estimate to the mean value of the observed response (as a percentage) is a measure of reproducibility of the model, and as a general rule, a model can be considered reasonably reproducible if its CV is not greater than 10 % (Ahmad et al. 2007; Ahmadi et al. 2005). The CV values obtained for all the responses studied were relatively small, none being higher than 3.57 %, as shown in Table 3. The  $R^2$  coefficient gives



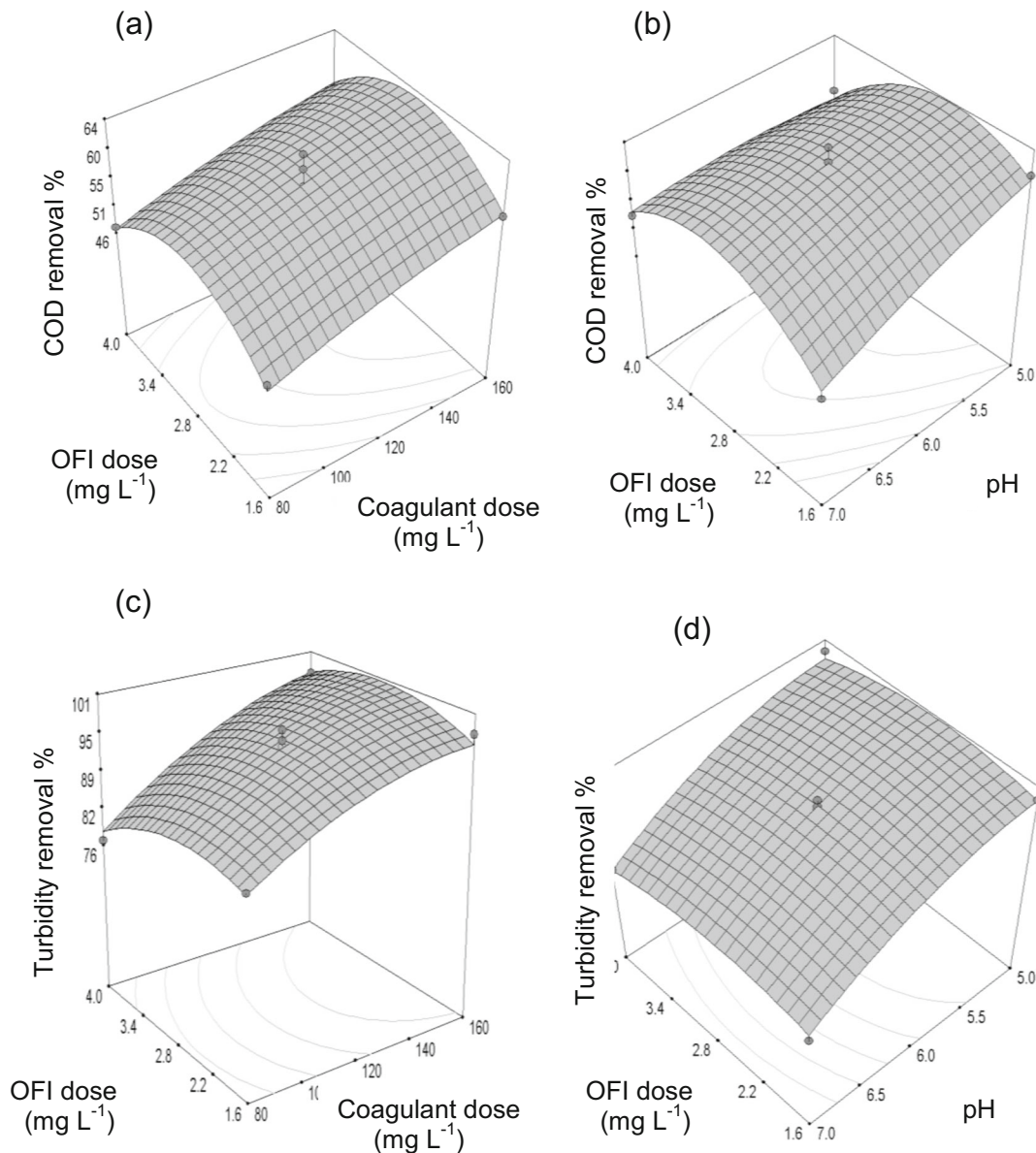
**Fig. 3** Correlation between the values predicted by the model and the experimental responses **a** percent COD removal and **b** percent turbidity removal for JL and **c** percent COD removal and **d** percent turbidity removal for FDM



the proportion of the total variation in the response predicted by the model, indicating ratio of sum of squares due to regression to total sum of squares. A high  $R^2$  value, close to 1, is desirable. A high  $R^2$  coefficient ensures a satisfactory adjustment of the quadratic model to the experimental data. The values of the coefficients of correlation ( $R^2$ ) were higher than 0.93, which shows a good fitting of the model for the responses, with low residual values (Fig. 3). According to Fig. 3, the

distributed points relatively near to the straight line show a good agreement between the predicted and observed values within the range of experiment (Ghafari et al. 2009).

A quadratic regression model was obtained for the four responses. Equations 3 and 6 give the coded values for the quadratic models. The independent variables for the models were coagulant dosage, adjuvant coagulant dosage, and pH, coded as  $x_1$ ,  $x_2$ , and  $x_3$ , respectively.



**Fig. 4** Three-dimension surface graphs for interactions  $X_1$ – $X_2$  (a),  $X_2$ – $X_3$  (b) for percent COD removal, and interactions between  $X_1$ – $X_2$  (c),  $X_2$ – $X_3$  (d) for percent turbidity removal for JL

*JL*COD removal  $Y_1$  (%)=

$$58.28 + 4.55x_1 - 0.018x_2 - 3.26x_3 + 4.40x_1x_3 \quad (3)$$

$$+ 2.95x_2x_3 - 7.22(x_2)^2$$

Turbidity removal  $Y_2$  (%)=

$$95.37 + 8.10x_1 - 0.90x_2 - 10.91x_3 + 6.10x_1x_3 - 2.88(x_1)^2 \quad (4)$$

$$- 3.73(x_2)^2 - 4.85(x_3)^2$$

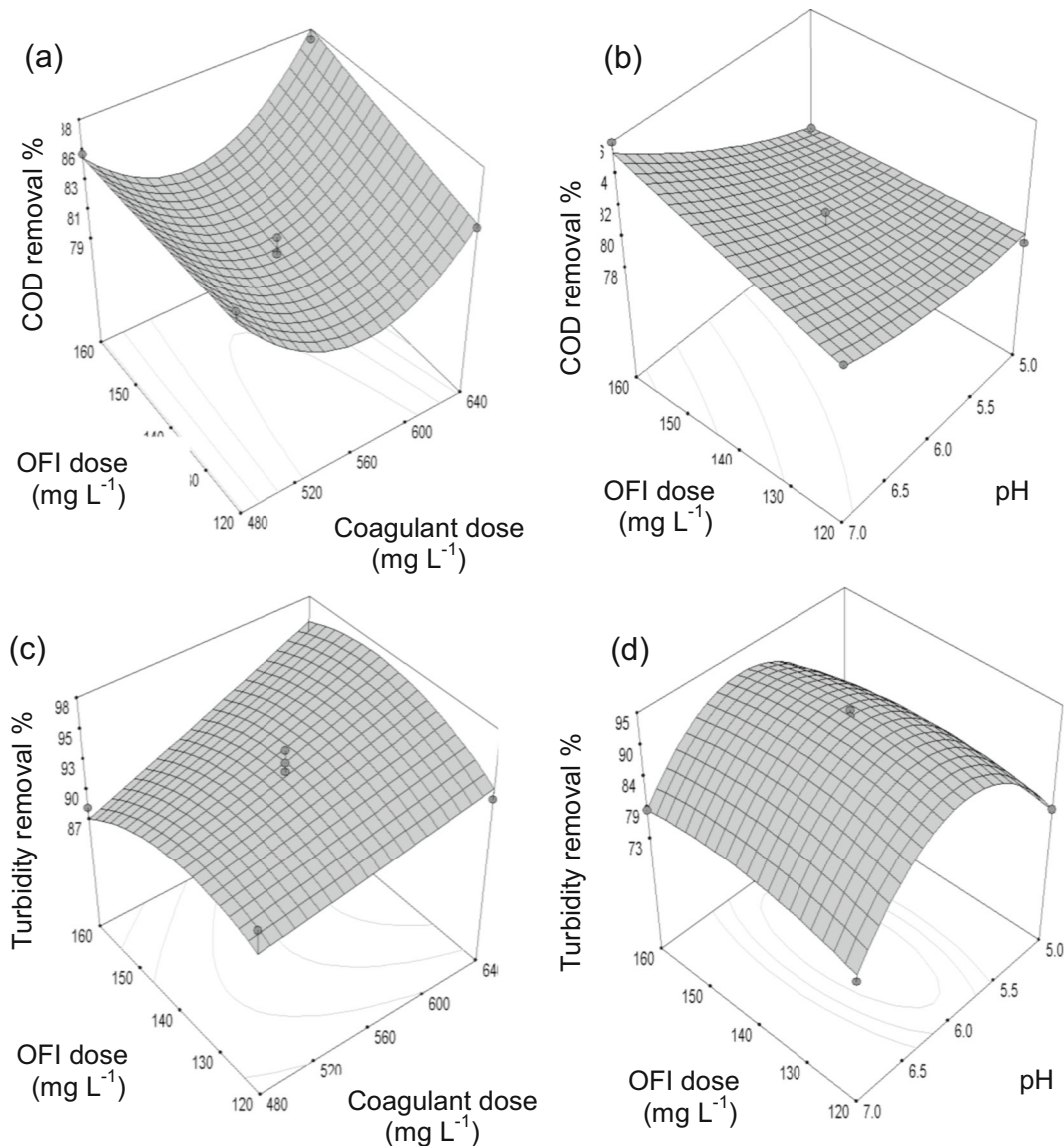
*FDM*COD removal  $Z_1$  (%)=

$$80.00 + 0.34x_1 + 1.04x_2 + 2.41x_3 + 1.07x_1x_2 \quad (5)$$

$$- 1.28x_1x_3 + 1.34x_2x_3 + 5.46(x_1)^2$$

Turbidity removal  $Z_2$  (%)=

$$92.42 + 2.95x_1 + 0.69x_3 - 15.04(x_3)^2 \quad (6)$$



**Fig. 5** Three-dimension surface graphs for interactions,  $X_1$ – $X_2$  (a),  $X_2$ – $X_3$  (b) for percent COD removal, and interactions between  $X_1$ – $X_2$  (c),  $X_2$ – $X_3$  (d) for percent turbidity removal for FDM



The coefficients with one factor represent the effect of the particular factor, while the coefficients with two factors and those with second-order terms represent the interaction between the two factors and quadratic effect, respectively. The positive sign in front of the terms indicates synergistic effect, while negative sign indicates antagonistic effect (Bhatia et al. 2007).

#### Analysis of the response surface for JL

For the graphical interpretation of the interactions, the use of three-dimensional plots of the regression model is highly recommended (Ahmadi et al. 2005).

The three-dimensional response surfaces of the interactions under study are presented in Fig. 4. These graphs are a function of two factors, with third factor kept at the central point.

Figure 4a shows that the concentration of  $\text{FeCl}_3$  is at its maximum level, that is,  $160.0 \text{ mg L}^{-1}$ , and that the concentration of OFI is close to its central point, between  $2.20$  and  $2.8 \text{ mg L}^{-1}$ , for which the removal of COD removal was the greatest. In Fig. 4b, the interactions of OFI and pH had the best responses when the pH was acid, that is, pH 5.0.

Relative to response percent removal of turbidity, Figure 4c, d shows that the best responses were obtained with maximum OFI and  $\text{FeCl}_3$  concentrations and minimum pH.

Figure 4a, b was concave for the model responses for the OFI concentration, which was due to the quadratic significance of the OFI concentration term for the COD

removal response. One can see that for coagulation aid concentrations higher than  $2.8 \text{ mg L}^{-1}$ , the response percent removal of COD (sum of suspended and dissolved organic matter) tends to decrease. This was probably due to the fact that the OFI extract increases the organic matter (dissolved fraction) of the system. After the coagulation process, part of the OFI organic matter left over remained in the supernatant, thus contributing to the final concentration of COD (Zhang et al. 2006). One must bear in mind that this increase results from a highly degradable vegetable species (Yin 2010).

#### Analysis of the response surfaces for FDM

According to Fig. 5a, the COD removal response decreased when the coagulant concentration was closer to its central point,  $560 \text{ mg L}^{-1}$ . The analysis of variables coagulation aid concentration and pH shows that the best responses for COD removal occurred when both variables were maximum,  $[\text{OFI}] = 640 \text{ mg L}^{-1}$  and pH 7.0, as can be seen in Fig. 5b. In Fig. 5b, the response percentage turbidity removal has a concave curvature towards the pH axis. This behavior is due to the high significance of the pH quadratic term. One can also see that the response decreased when the pH was above its central point. Now, for the coagulant and coagulation aid concentrations, Fig. 5a, b, the best responses occurred when the concentration of  $\text{FeCl}_3$  was maximum,  $640 \text{ mg L}^{-1}$  and the OFI concentration was between  $130$  and  $150 \text{ mg L}^{-1}$ .

**Table 4** Comparison between the values of COD and turbidity for textile effluents *in natura*, and the responses of the predicted values and experimentally obtained after coagulation of the effluents from a jeans laundry (JL) and fabric dyeing mill (FDM)

|                 | COD ( $\text{mg L}^{-1}$ ) |         |                     | Turbidity (FTU)  |         |                   |
|-----------------|----------------------------|---------|---------------------|------------------|---------|-------------------|
|                 | <i>In natura</i>           | Predict | Actual              | <i>In natura</i> | Predict | Actual            |
| JL              | $1,094.20 \pm 0.0020$      |         |                     | $104 \pm 1.4$    |         |                   |
| OP <sup>a</sup> |                            | 394.24  | $385.49 \pm 0.5668$ |                  | 9.42    | $9.09 \pm 0.2312$ |
| CV (%)          |                            |         | 0.88                |                  |         | 0.25              |
| FDM             | $1,264.21 \pm 0.2312$      |         |                     | $31.5 \pm 0.71$  |         |                   |
| OP <sup>b</sup> |                            | 147.66  | $161.94 \pm 0.8000$ |                  |         | $2.01 \pm 0.499$  |
| CV (%)          |                            |         | 0.91                |                  | 1.34    | 1.58              |

CV (%) coefficient of variation (%)

<sup>a</sup> OP optimum point ( $[\text{FeCl}_3] = 160 \text{ mg L}^{-1}$ ,  $[\text{OFI}] = 2.60 \text{ mg L}^{-1}$  e pH 5.0)

<sup>b</sup> OP optimum point ( $[\text{FeCl}_3] = 640 \text{ mg L}^{-1}$ ,  $[\text{OFI}] = 160 \text{ mg L}^{-1}$  e pH 6.0)

The analysis of Figs. 4 and 5 shows that the concentration of OFI for FDM was not as significant as for JL. It is worth noting that actual textile effluents undergo distinct processes and therefore have different composition and physical chemical parameters. Thus, they will respond in a rather different way in an optimization process.

### Process optimization

The coagulation/flocculation process was carried out in optimal conditions to maximize the COD and turbidity removal responses for the effluents under study. After evaluating the quadratic model and the model of the response surface, the software proposed various assays was selected taking into consideration the maximum DQO and turbidity removal. The best conditions obtained for the JL variables were the following:  $[\text{FeCl}_3] = 160 \text{ mg L}^{-1}$ ,  $[\text{OFI}] = 2.60 \text{ mg L}^{-1}$ , and pH 5.0. The FDM optimized conditions were the following:  $[\text{FeCl}_3] = 640 \text{ mg L}^{-1}$ ,  $[\text{OFI}] = 160 \text{ mg L}^{-1}$ , and pH 6.0 (Table 4).

According to the results, the proposed method offered good reproducibility with low values of standard deviation and variation coefficients.

### Conclusions

The saline extraction process was efficient in the isolation of the coagulation activity of OFI. To evaluate the significance of the natural coagulation aid under treatment, the effluents were optimized using the RSM. The data were adjusted to a quadratic model with variance analysis (ANOVA), which showed that the OFI concentration was significant for the percent COD removal and the percent turbidity removal responses for both effluents under study. In the optimum conditions determined for JL obtained as percentage removal of COD and turbidity:  $64.77 \% \pm 0.57$  and  $91.26 \% \pm 0.23$ , respectively. For the FDM obtained as percentage removal of COD and turbidity:  $87.19 \% \pm 0.80$  and  $93.62 \% \pm 1.50$ , respectively.

OFI has significant coagulant activity and may be an efficient alternative for the treatment of textile effluents.

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